

The question

When contacts-v1-exp199-1.5B emits a contact set in one rollout, is that set jointly realizable as a single 3D structure — more so than a contact set carrying the same per-pair marginals but assembled across different rollouts?

Plainly: does autoregressive generation produce a coherent structural hypothesis, or a bag of independently-drawn contacts that happen to share one document?

Issue #211 — github.com/Open-Athena/MarinFold/issues/211

Why it is worth asking

Every contacts-v1 eval we have (#82, #89, #180) collapses the 100 rollouts per protein into per-pair vote counts and scores the marginals. The per-rollout partition is discarded at the moment of the increment, so nothing has ever measured whether a single rollout is internally coherent.

Two prior results say the joint is where the information is, and neither tests generation.
#201 Phase 0: 77% of the val loss is nuisance permutation entropy, so next-token CE barely scores the joint. #163: conditioning on TRUE partial contact maps lifts R-precision 0.145 to 0.556.

If consistency ranks rollouts without ground truth it is also a best-of-N selector and a candidate RL reward for #200 / #208 — usable on sequences with no known structure, which is the entire ESM-Atlas half of #199's training mixture.

The method that did not work

The natural approach — Floyd-Warshall bound smoothing over the distances implied by the contacts plus the 3.8 Å CA(i)-CA(i+1) bond — has no discriminative power. It reports zero violations for a true contact set AND zero for a separation-matched random one, at four different bound pairs.

The reason is structural, not a tuning failure. A violation needs a path of upper bounds summing below some lower bound. Any two-hop contact path is $2U$, about 20-24 Å, and a contact plus k backbone steps is $U + 3.8k$ — so nothing reaches under a ~ 10 Å lower bound once $\text{min_seq_separation} = 6$ has excluded the close-in-sequence pairs.

Triangle smoothing tests feasibility in an arbitrary METRIC SPACE, which a contact graph satisfies almost for free. It does not test feasibility in R^3 .

What works instead

The 3D embedding residual: minimise bond, contact, non-contact and steric violation over x in $R^{(L \times 3)}$, and report the leftover contact excess, minimised over restarts.

Floyd-Warshall is kept as step 1 of 3 — it is the bound-smoothing step of the classic Crippen-Havel EMBED algorithm and the right preconditioner for the embedding — but it is not the metric.

Bounds are measured on all 554 ground-truth proteins, not assumed (91,525 contact pairs, 11.1M non-contact pairs), reusing exp174's published GT bundle:

bond = 3.809 Å (median CA(i)-CA(i+1))

u_contact = 13.02 Å (contact CA-CA p99.5)

l_noncontact = 6.46 Å (non-contact CA-CA p0.5)

d_min = 3.53 Å (closest non-bonded CA pair, p1)

The number that governs every claim

10.7% of non-contact pairs sit closer than u_{contact} .

pyconfind contacts are SIDE-CHAIN contacts, so CA-CA distance is only a proxy for them and no threshold pair separates the two populations. A nonzero residual therefore means "less geometrically consistent", never "provably unrealizable".

The arm comparison is unaffected: every arm is scored under identical bounds, so the scale cancels. But the absolute number is not a proof of anything, and the writeup does not treat it as one.

The calibration gate passed, and corrected the issue

Ground truth beats a separation-matched random set of the same size and the same $|i-j|$ profile by 5.6x in median per-contact excess, on 89.6% of the 470 chain-break-free proteins — 95.4% at L 100-200, 88.3% at L 200-350, and 100% at L ≥ 350 .

Two of the three criteria written into the issue were wrong. "GT scores about 0" is incoherent against the bounds it was paired with: u_{contact} is a p99.5, so $\sim 0.5\%$ of real contacts exceed it by construction and ground truth carries a structural nonzero floor. The gate is now relative.

The decoy arm is not a floor. A different real protein's contact map scores the same as the true one (0.0384 vs 0.0337; truth wins on 49.6%, a coin flip). That is correct — the score is sequence-blind — but it bounds the claim: this experiment detects "not a fold at all", not "wrong fold copied from a real one".

Scope limit: below L = 100 the metric is near-blind (GT 0.0000 vs random 0.0011) — a short chain embeds almost anything. A further 15% of the eval set has a chain break and is not embeddable as the continuous chain a contacts-v1 document asserts. Both subsets are reported separately, never silently dropped.

How sensitive is it where the model actually lives?

The gate contrast is easy. The real contrast is a rollout against a chimera built from the same rollouts with the same marginals — both roughly 60/40 true/false mixtures of the same contacts. Sweeping corruption across #199's operating band (R-precision about 0.59, so about 40% wrong), over 60 proteins at $L \geq 100$:

a 0.05 step in corruption: sign consistency 53-65%, p 0.04-0.35 — NOT reliably separable at $n = 60$.

a 0.10 step: sign consistency 58.3%, $p = 0.006$ — separable.

a 0.20 step: sign consistency 73.3%, $p = 2.3e-05$ — strongly separable.

So the resolvable effect is bounded and the experiment's power has to come from scale. This sweep is also an UPPER bound on the real effect: the chimera keeps every pair the model proposed and only breaks their co-occurrence, a gentler perturbation than swapping pairs for random ones.

The arms

Every arm is size-matched. #142 measured rollouts emitting about 0.70x the ground-truth contact count, and a sparser set is trivially easier to embed, so an unmatched comparison would read under-generation as consistency.

ground truth — calibration ceiling. GT size-matched — removes the count confound.

rollout — the treatment.

marginal-matched chimera — THE KEY NULL: pairs drawn from the pooled rollout votes with probability proportional to vote count.

splice chimera — half of rollout a plus half of rollout b. Decoy protein — sequence-blindness ceiling. Separation-matched random — floor.

Why the marginal chimera is the sharp test: it shares the model, the protein, the per-pair marginals and the set size with the rollout. The only difference is whether the contacts were drawn jointly in one autoregressive pass or independently from the pooled vote distribution. Any gap is joint structure the model put there while generating, and cannot be explained by marginal accuracy — which is all any existing eval measures.

Results

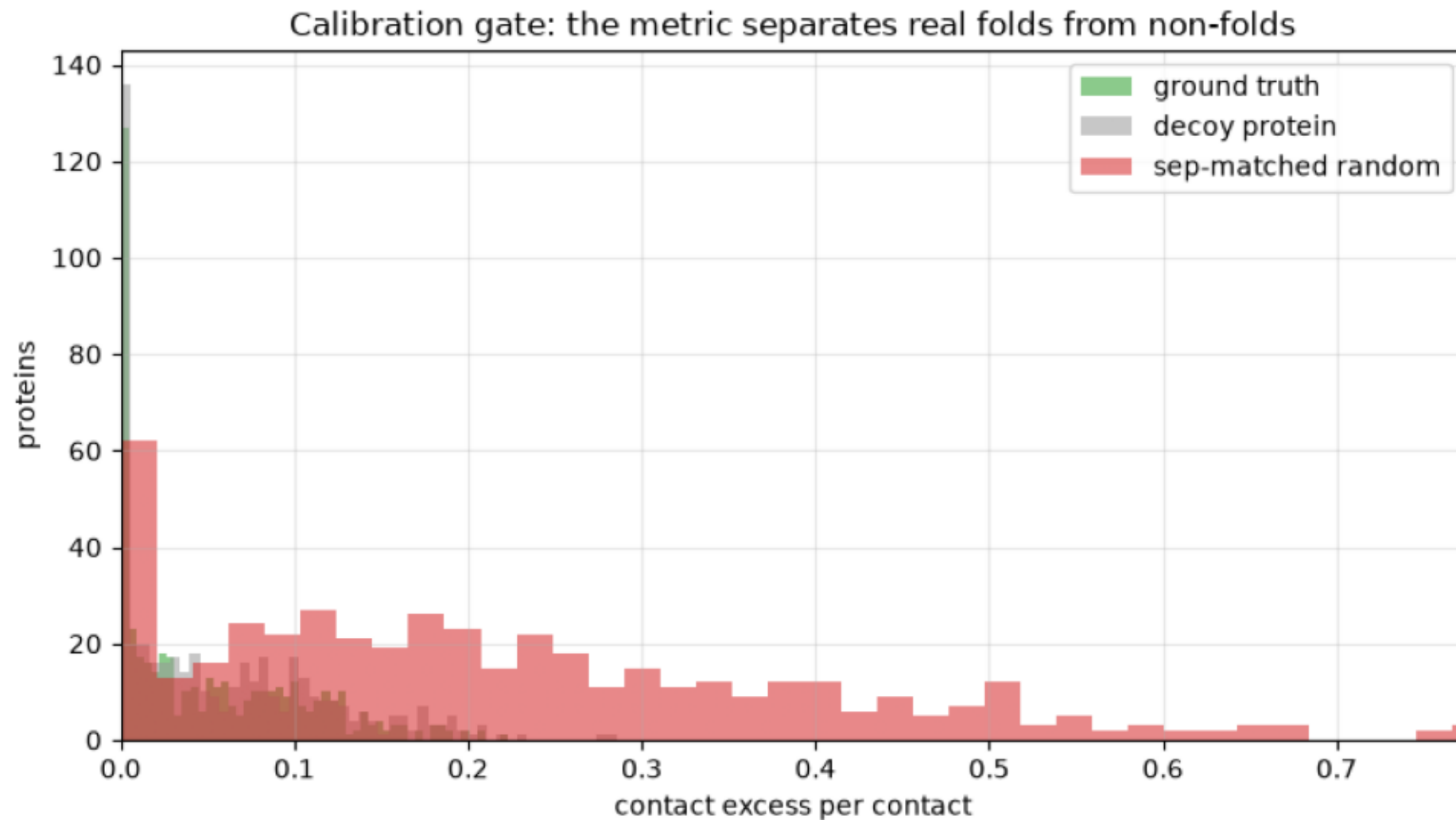
(Filled in by analyze.py once scoring completes.)

Conclusion

(Filled in once the primary contrast is measured.)

calibration_gate.png

Ground truth sits 5.6× below random. A decoy protein ties with the truth — the score is sequence-blind.



dose_response.png

Score vs corruption, 95% CI. Grey band marks where #199 actually generates (R-precision ≈ 0.59).

